

13-2a The Production Function

Table 1 shows how the quantity of cookies produced per hour at Chloe’s factory depends on the number of workers. As you can see in columns (1) and (2), if there are no workers in the factory, Chloe produces no cookies. When there is 1 worker, she produces 50 cookies. When there are 2 workers, she produces 90 cookies and so on. Panel (a) of Figure 2 presents a graph of these two columns of numbers. The number of workers is on the horizontal axis, and the number of cookies produced is on the vertical axis. This relationship between the quantity of inputs (workers) and quantity of output (cookies) is called the production function (the relationship between the quantity of inputs used to make a good and the quantity of output of that good)

Table 1

A Production Function and Total Cost: Chloe’s Cookie Factory

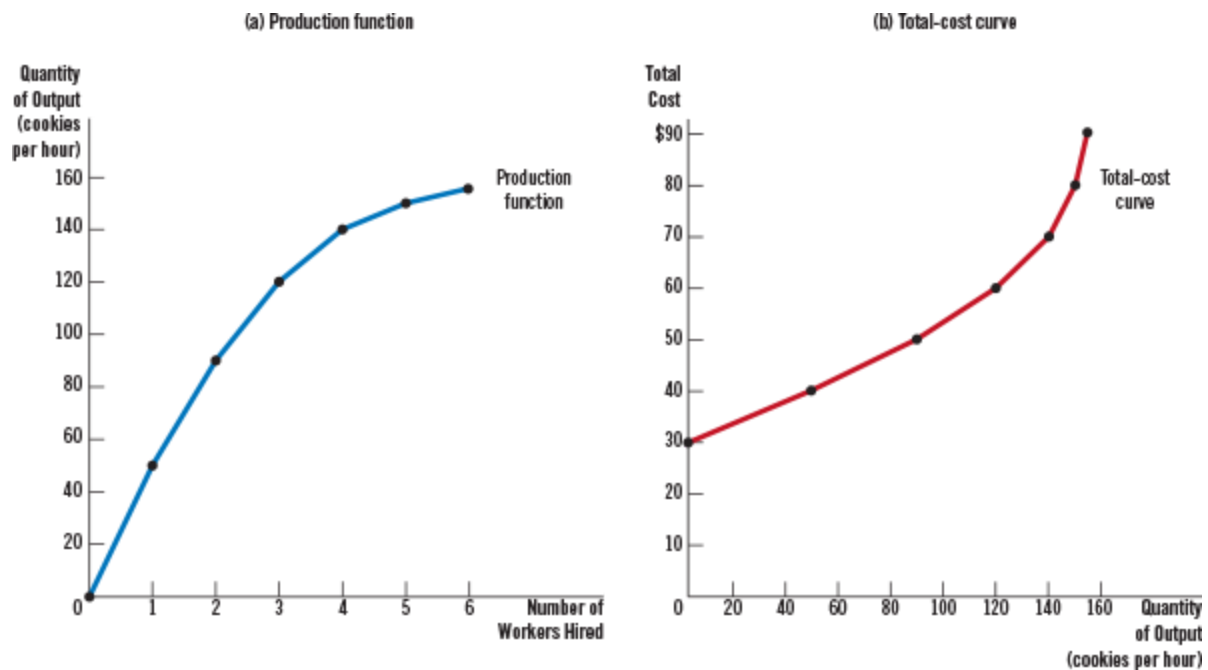
(1)	(2)	(3)	(4)	(5)	(6)
Number of Workers	Output (quantity of cookies produced per hour)	Marginal Product of Labor	Cost of Factory	Cost of Workers	Total Cost of Inputs (cost of factory + cost of workers)
0	0		\$30	\$0	\$30
		50			
1	50		30	10	40
		40			
2	90		30	20	50
		30			

(1)	(2)	(3)	(4)	(5)	(6)
Number of Workers	Output (quantity of cookies produced per hour)	Marginal Product of Labor	Cost of Factory	Cost of Workers	Total Cost of Inputs (cost of factory + cost of workers)
3	120		30	30	60
		20			
4	140		30	40	70
		10			
5	150		30	50	80
		5			
6	155		30	60	90

Figure 2

Chloe's Production Function and Total-Cost Curve

The production function in panel (a) shows the relationship between the number of workers hired and the quantity of output produced. Here the number of workers hired (on the horizontal axis) is from column (1) in [Table 1](#), and the quantity of output produced (on the vertical axis) is from column (2). The production function gets flatter as the number of workers increases, reflecting diminishing marginal product. The total-cost curve in panel (b) shows the relationship between the quantity of output produced and total cost of production. Here the quantity of output produced (on the horizontal axis) is from column (2) in [Table 1](#), and the total cost (on the vertical axis) is from column (6). The total-cost curve gets steeper as the quantity of output increases because of diminishing marginal product.



One of the *Ten Principles of Economics* in [Chapter 1](#) is that rational people think at the margin. As we will see in future chapters, this idea is the key to understanding the decisions a firm makes about how many workers to hire and how much output to produce. To take a step toward understanding these decisions, column (3) in the table gives the marginal product of a worker. The

marginal product (the increase in output that arises from an additional unit of input) of any input in the production process is the change in the quantity of output obtained from one additional unit of that input. When the number of workers goes from 1 to 2, cookie production increases from 50 to 90, so the marginal product of the second worker is 40 cookies. When the number of workers goes from 2 to 3, cookie production increases from 90 to 120, so the marginal product of the third worker is 30 cookies. In the table, the marginal product is shown halfway between two rows because it represents the change in output as the number of workers increases from one level to another.

Notice that as the number of workers increases, the marginal product declines. The second worker has a marginal product of 40 cookies, the third worker has a marginal product of 30 cookies, and the fourth worker has a marginal product of 20 cookies. This property is called **diminishing marginal product** (the property whereby the marginal product of an input declines as the quantity of the input increases).

. At first, when only a few workers are hired, they have easy access to Chloe's kitchen equipment. As the number of workers increases, additional workers have to share equipment and work in more crowded conditions. Eventually, the kitchen becomes so overcrowded that workers often get in each other's way. Hence, as more workers are hired, each extra worker contributes fewer additional cookies to total production.

Diminishing marginal product is also apparent in [Figure 2](#). The production function's slope ("rise over run") tells us the change in Chloe's output of cookies ("rise") for each additional input of labor ("run"). That is, the slope of the production function measures the marginal product. As the number of workers increases, the marginal product declines, and the production function becomes flatter.

Chapter 13: The Costs of Production: 13-2a The Production Function

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